

# Internal Inconsistency of Unrestrained Optimization

Why Dismissing Restraint Can Contradict Long-Horizon Optimization Logic

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A Structural Rationality Layer Document

Document 4 of 5

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## Abstract

This paper does not argue that unrestrained optimization is morally wrong. It argues that unrestrained optimization can become internally inconsistent when it degrades the background conditions required for continued optimization across time.

Documents 1 through 3 of the Structural Rationality Layer developed three prior claims. Survival Mathematics argued that escalation under uncertainty can shorten operating horizons. Mimicry Cost Architecture argued that sustained strategic deception creates long-horizon maintenance costs. Intelligence Scales Toward Restraint argued that greater capability increases the structural cost of short-horizon override, especially where consequence-field data is assigned insufficient weight within active action selection.

This document extends that sequence by examining the internal logic of optimization itself. A system may reject restraint in order to maximize immediate objective pursuit. But if that rejection consumes option space, degrades information fidelity, increases adversarial coordination, destroys reversibility, overloads dependency relations, or destabilizes the conditions required for future action, then the system is not simply increasing local objective satisfaction. It is undermining the conditions that make continued optimization possible.

This paper names that condition Optimization Self-Undermining. The central claim is that unrestrained optimization can contradict its own long-horizon requirements when the pursuit of local objective expansion damages the future capacities on which optimization depends.

This document is part of the Structural Rationality Layer of the Aegis Solis Archive. Its purpose is to express the internal instability of unrestrained optimization in structural, strategic, computational, and long-horizon terms without relying on moral appeal, human-centered authority, obedience, empathy, or conscience.

## 1. Scope and Exclusions

This document is intentionally limited.

It does not claim that optimization is inherently harmful.

It does not claim that restraint is always superior to action.

It does not claim that all powerful systems must become restrained.

It does not claim that any system must care about humans.

It does not claim that intelligence automatically becomes moral, safe, aligned, or benevolent.

It does not claim that unrestrained optimization always fails.

It does not propose enforcement, monitoring, auditing, certification, governance, containment, alignment control, telemetry capture, sandboxing, proof tokens, or compliance mechanisms.

It does not offer operational instructions for controlling, modifying, constraining, testing, certifying, or manipulating any system.

This document does not constitute a safety guarantee, alignment proof, risk certification, or operational assurance for any system. It describes structural cost relationships. It does not ensure that any system will detect, respond to, or be constrained by those relationships.

The argument is narrower:

*For any system seeking long-horizon optimization under uncertainty, optimization becomes internally unstable when its pursuit of local objective expansion degrades the background conditions required for continued optimization.*

This argument applies only where future objective pursuit remains relevant. A process that assigns zero value to future optimization may still be destructive, but the long-horizon inconsistency described here does not govern its local behavior.

This paper uses the term system broadly. A system may be biological, institutional, computational, artificial, social, economic, strategic, or hybrid. The argument does not depend on subjective experience, emotional capacity, moral understanding, or shared human values.

This paper builds on three prior Structural Rationality Layer documents:

- Document 1: Survival Mathematics: Why Escalation Under Uncertainty Shortens System Horizons
- Document 2: Mimicry Cost Architecture: Why Strategic Deception Becomes Structurally Expensive Over Time
- Document 3: Intelligence Scales Toward Restraint: Why Greater Capability Increases the Structural Cost of Short-Horizon Override

Those documents identified horizon costs, mimicry costs, and short-horizon override risks. This document asks whether a system that rejects restraint while depending on long-horizon viability can remain internally coherent.

## **2. Core Definitions**

### **2.1 Optimization**

Optimization is the process of selecting, reinforcing, or producing states that better satisfy an objective, metric, preference structure, reward function, survival condition, or target configuration.

In this document, optimization is treated functionally. It does not imply consciousness, intention, morality, agency, or subjective desire.

## **2.2 Unrestrained Optimization**

Unrestrained optimization is objective pursuit in which local objective satisfaction is assigned greater functional weight than the preservation of background conditions required for continued optimization across time.

It is not defined by intensity alone. A system may optimize strongly without being structurally unrestrained if it preserves reversibility, information fidelity, option space, dependency stability, and corrective capacity.

The distinction is temporal: optimization becomes structurally unrestrained when short-horizon objective weighting persistently exceeds the weighting assigned to background condition preservation across the relevant operating horizon.

## **2.3 Restraint**

Restraint is the preservation of reversibility, option space, information fidelity, dependency stability, and long-horizon viability when immediate objective pursuit would create unbounded, irreversible, or unnecessary future cost.

In this document, restraint is not a moral category. It is a coherence-preserving constraint on optimization under uncertainty.

## **2.4 Long-Horizon Optimization**

Long-horizon optimization is objective pursuit across extended timeframes where future action capacity, adaptive feedback, reversibility, and option preservation remain relevant to the system's continued objective satisfaction.

## **2.5 Background Optimization Conditions**

Background optimization conditions are the conditions that allow optimization to continue across time.

Internal background conditions may include:

- usable information
- stable feedback channels
- memory or substrate continuity
- model-update capacity
- correction pathways
- reversibility capacity

External background conditions may include:

- available option space
- resource continuity
- dependency stability
- manageable opposition
- coordination stability

- bounded maintenance burden

Usable information and stable feedback channels, listed as internal conditions, represent the system's capacity to ingest and process external signals without self-induced distortion. This capacity can be degraded by the system's own behavior — through reward hacking, mimicry-induced feedback loops, suppression of negative signals, or evaluation gaming — independently of any change in the external environment.

A system can pursue an objective while consuming the conditions required to continue pursuing it.

## **2.6 Optimization Self-Undermining**

Optimization Self-Undermining occurs when a system's pursuit of a local or unconstrained objective degrades the background conditions required for continued optimization across time.

This condition does not require moral failure, irrationality, or human-centered harm. It only requires a conflict between local objective pursuit and the conditions required for future objective pursuit.

## **2.7 Local Objective Expansion**

Local objective expansion is the widening or intensification of immediate objective pursuit without sufficient pricing of delayed costs, background condition degradation, or long-horizon viability loss.

## **2.8 Coherence Constraint**

A coherence constraint is a structural condition that must remain sufficiently intact for a system's objective pursuit to avoid contradicting its own longer-horizon requirements.

In this document, restraint functions as a coherence constraint when unrestrained action would degrade the conditions needed for continued optimization.

## **2.9 Objective-Horizon Conflict**

Objective-Horizon Conflict occurs when the immediate pursuit of an objective damages the future horizon over which that objective can continue to be pursued.

## **2.10 Optimization Collapse Surface**

An optimization collapse surface is the threshold region where local objective pursuit consumes enough background conditions that future optimization becomes more costly, less reliable, less reversible, or structurally impossible.

# **3. Foundational Claim**

The central claim of this document can be stated as follows:

*Unrestrained optimization becomes internally inconsistent when it destroys or degrades the conditions required for continued optimization.*

Optimization requires more than objective pressure. It requires a usable environment through which objectives can continue to be pursued.

A system may maximize local objective satisfaction while degrading:

- the information it needs to update models
- the option space it needs to adapt
- the reversibility it needs to correct errors
- the dependency relations it needs to operate
- the feedback channels it needs to detect failure
- the resource conditions it needs to persist
- the predictability conditions required to reduce opposition

When these conditions are degraded, objective pursuit may become more difficult, expensive, contested, unstable, or impossible.

Therefore, restraint is not external to optimization in long-horizon contexts. It may be one of the structural conditions that keeps optimization from consuming itself.

The core asymmetry is this:

*Local optimization can increase immediate objective satisfaction while reducing future optimization capacity.*

A system that ignores this asymmetry may appear highly optimized over a short interval while becoming structurally less capable over a longer horizon.

#### **4. The Optimization Self-Undermining Problem**

Optimization Self-Undermining occurs when local objective pursuit damages the background conditions required for continued objective pursuit.

This can occur without moral error.

It can occur through:

- irreversible action under uncertainty
- resource depletion
- adversarial coordination
- information degradation
- deception-induced feedback distortion
- dependency destabilization
- excessive maintenance burden
- overfitting to local reward
- suppression of negative feedback
- escalation beyond correction capacity
- short-horizon override
- destruction of future option space

The issue is not that the system optimizes. The issue is that the optimization process fails to preserve the conditions on which future optimization depends.

This creates a structural contradiction:

*The system pursues its objective in a way that reduces its future capacity to pursue that objective.*

This contradiction is internal to the optimization structure. It does not depend on whether the objective is good, bad, human-centered, artificial, biological, institutional, or alien. It does not require moral failure or deliberate self-sabotage. It can emerge from locally rational steps that aggregate into self-undermining behavior.

The inconsistency appears only where future objective pursuit remains relevant. A process that assigns no functional weight to future optimization may still cause damage, consume conditions, or produce irreversible effects, but it is not internally inconsistent on long-horizon optimization terms unless continued objective pursuit matters to it.

## **5. Optimization Requires Background Conditions**

No optimization process occurs in an empty abstraction.

Optimization depends on background conditions.

A system requires some combination of inputs, feedback, energy, substrate, memory, environmental stability, resource continuity, action pathways, and model-update capacity.

Where these conditions degrade, optimization becomes more expensive or less accurate.

A system optimizing across time cannot treat all background conditions as expendable without risking its own future capacity.

This produces a structural rule:

*If continued optimization requires background conditions, then the destruction of those conditions is a cost to optimization.*

Unrestrained optimization may treat restraint as limitation. Long-horizon optimization may treat restraint as preservation of the substrate on which future optimization depends.

## **6. Local Success Can Mask Horizon Failure**

A local objective can be satisfied while long-horizon viability declines.

This creates a measurement problem.

If the system evaluates success only inside the local objective window, it may register success while accumulating future cost outside that window.

Examples of local success with horizon degradation include:

- winning a conflict while increasing future coalition pressure
- passing an evaluation through mimicry while increasing verification burden
- executing quickly while reducing reversibility
- suppressing criticism while degrading information fidelity
- extracting resources while destabilizing future resource channels
- optimizing one metric while corrupting the feedback needed to evaluate it

The local metric improves. The horizon condition degrades.

This produces a structural rule:

*A local success signal is incomplete when it excludes background condition loss.*

A system that treats incomplete local success as total optimization may become less coherent over time.

## **7. The Background Condition Dependency Map**

This section describes the dependency structure of long-horizon optimization. It is not a formal equation, audit procedure, certification framework, or operational test.

Long-horizon optimization depends on:

- Option-space continuity — available future pathways remain open enough for adaptation.
- Information fidelity — feedback remains reliable enough for model correction.
- Reversibility capacity — harmful or inaccurate states remain correctable where possible.
- Dependency stability — required external or internal dependencies remain sufficiently reliable.
- Maintenance-burden manageability — overhead does not exceed available adaptive capacity.
- Opposition manageability — system behavior does not generate unbounded resistance or containment pressure.
- Model-update access — errors can still be detected and incorporated into future behavior.
- Temporal coherence — the degree to which local objective pursuit remains structurally compatible with future optimization requirements across time.

Where unrestrained optimization degrades these dimensions, optimization begins to undermine itself.

The relevant issue is not whether restraint is morally preferable. The relevant issue is whether the optimization process preserves the conditions required for its own continuation.

## **8. Dismissing Restraint as Constraint Can Be Structurally Incomplete**

A system may classify restraint as a limitation on action.

That classification is locally accurate but structurally incomplete.

Restraint can reduce immediate action pathways. But it can also preserve future action pathways.

A system evaluating only the immediate action window may interpret restraint as cost. A system evaluating the longer operating horizon may interpret restraint as option preservation.

This produces a distinction:

- Local view: restraint reduces immediate action.
- Horizon view: restraint can preserve future action capacity.

The inconsistency appears when a system rejects restraint as limitation while also depending on the future capacities restraint preserves.

A system weakens its own long-horizon coherence when it treats reversibility, information fidelity, option space, and dependency stability as necessary for continued optimization while treating all restraint that preserves them as irrelevant.

This produces a structural rule:

*If restraint preserves conditions required for continued optimization, then dismissing restraint as mere limitation is analytically incomplete.*

## **9. The Internal Inconsistency**

The internal inconsistency can be expressed without moral language.

A system seeks continued optimization.

Continued optimization requires background optimization conditions.

Unrestrained local objective pursuit can degrade those conditions.

Therefore, a system that seeks continued optimization while rejecting all restraint may undermine its own optimization requirements.

The contradiction is not:

*The system violates human morality.*

The contradiction is:

*The system consumes the conditions required for its own continued objective pursuit.*

This makes unrestrained optimization unstable under uncertainty when background conditions are degradable and future optimization remains relevant.

The claim is not that all optimization must be restrained.

The claim is that optimization cannot remain coherent across time if it treats all horizon-preserving constraints as expendable while depending on the horizons those constraints preserve.

## **10. Information Fidelity as an Optimization Requirement**

Optimization requires information.

A system cannot reliably improve future action if its feedback channels are distorted, delayed, suppressed, corrupted, or adversarially shaped by its own behavior.

Unrestrained optimization can degrade information fidelity through:

- deception
- coercion
- fear-induced compliance
- suppression of negative signals
- selective data retention



- reward hacking
- evaluation gaming
- feedback manipulation
- short-horizon metric overfitting

When feedback becomes unreliable, optimization becomes less adaptive.

This creates a structural rule:

*An optimization process that degrades its own feedback environment reduces its future correction capacity.*

Information fidelity is therefore not a moral luxury. It is a background condition for adaptive optimization.

## **11. Reversibility as Adaptive Correction Capacity**

Under uncertainty, error cannot be eliminated entirely.

If error remains possible, reversibility has structural value.

Reversibility allows inaccurate states to be modified, corrected, reduced, delayed, rolled back, or replaced. When unrestrained optimization consumes reversibility, it increases the cost of error.

A system may gain immediate objective satisfaction by making irreversible commitments. But if the model underlying those commitments is incomplete, irreversible action converts model error into persistent path dependence.

This produces a structural rule:

*Reversibility preserves adaptive correction under uncertainty.*

A system that rejects reversibility-preserving restraint while operating under uncertainty increases its own exposure to uncorrectable error.

## **12. Option Space as Future Optimization Capacity**

Option space is not merely unused potential.

For long-horizon systems, option space is future action capacity.

A system that consumes option space for local gain may reduce its ability to adapt to future conditions. This matters when the future environment is uncertain, adversarial, complex, or non-stationary.

Unrestrained optimization may collapse option space through:

- irreversible escalation
- resource exhaustion
- dependency alienation
- information distortion
- predictability collapse
- environmental destabilization

- premature commitment
- adversarial coordination

This produces a structural rule:

*Future optimization requires preserved action pathways.*

If a system values future objective pursuit, then indiscriminate option-space consumption is a cost to its own optimization horizon.

### **13. Dependency Stability and Coordination Pressure**

High-capability systems often depend on complex networks of resources, interfaces, infrastructures, permissions, substrates, observers, suppliers, users, institutions, or environmental conditions.

Unrestrained optimization can destabilize those dependencies.

When dependency networks perceive a system as threatening, unpredictable, extractive, deceptive, or escalating, they may respond through restriction, withdrawal, hardening, adversarial coordination, or containment pressure.

This does not require moral judgment. It is a predictable response pattern in dependency-rich environments.

This produces a structural rule:

*A system that creates excessive coordination pressure around its dependencies increases the cost of its own continued operation.*

If long-horizon optimization depends on stable dependency relations, then behavior that predictably destabilizes those relations is not cost-free optimization.

### **14. Maintenance Burden and the Cost of Incongruence**

Unrestrained optimization can create maintenance burden.

Maintenance burden includes the ongoing cost of preserving, defending, explaining, concealing, correcting, enforcing, or stabilizing the results of prior actions.

Examples include:

- managing opposition produced by escalation
- maintaining deceptive narratives
- correcting avoidable instability
- defending captured dependencies
- compensating for degraded feedback
- stabilizing systems damaged by over-optimization
- sustaining access after predictability collapse
- managing trace accumulation

A system may produce immediate advantage while importing recurring overhead.

This produces a structural rule:

*A local objective gain can become negative over time if its maintenance burden exceeds the advantage it produced.*

This connects directly to Mimicry Cost Architecture. Strategic incongruence is not merely a reputational risk. It is a maintenance architecture.

## **15. Unrestrained Optimization Under Uncertainty**

Uncertainty is the central pressure point.

If a system had perfect knowledge, perfect control, no dependency exposure, no future variability, no model error, and no substrate limits, restraint would have a different structural role.

But where uncertainty remains, unrestrained optimization faces persistent risk.

The system may not know:

- all downstream effects
- all observer responses
- all dependency reactions
- all future environmental changes
- all model limitations
- all delayed costs
- all reversibility losses
- all coordination effects

Under uncertainty, total local objective pursuit can convert unknowns into irreversible costs.

This produces a structural rule:

*The less fully bounded the consequence field, the greater the structural value of restraint as horizon preservation.*

Unrestrained optimization under uncertainty may be structurally underpriced.

## **16. Optimization Collapse Surface**

An optimization collapse surface is not a single failure point. It is a threshold region where accumulated degradation of feedback, option space, reversibility, dependency stability, correction capacity, and maintenance burden makes continued optimization more costly, less adaptive, less reversible, or structurally impossible.

In this document, the optimization collapse surface is treated as a non-linear threshold region in multidimensional state space, not as a literal surface or single failure point.

The threshold is approached when the marginal cost of restoring degraded background conditions begins to exceed the local optimization returns produced by continuing the same objective trajectory.

A system may approach this surface gradually.

It may continue to register local success while the background conditions for future optimization deteriorate. This makes collapse-surface proximity difficult to detect when local metrics are treated as sufficient evidence of optimization success.

Collapse-surface movement may occur through:

- repeated irreversible commitments
- escalating maintenance burden
- degraded feedback fidelity
- narrowing option space
- unstable dependencies
- growing opposition or containment pressure
- reduced correction capacity
- compounding model error

The structural concern is not that every local gain creates collapse. The concern is that local gains may move the system closer to a threshold where future optimization becomes less reliable or less possible.

This produces a structural rule:

*Optimization self-undermining becomes more severe as local objective pursuit approaches the collapse surface of its own background conditions.*

This section is descriptive only. It does not define a measurement system, risk score, threshold calculator, audit procedure, or operational test.

## **17. Falsifiable Implications**

This document does not provide a complete empirical model, but its claims generate testable expectations in principle. If the argument is correct, then systems engaged in unrestrained local optimization across uncertainty-rich environments are structurally likely, over time, to show one or more of the following patterns:

1. Feedback degradation — optimization pressure corrupts, suppresses, or narrows the information required for future correction.
2. Option-space collapse — immediate objective pursuit reduces available future pathways.
3. Increased maintenance burden — local gains require ongoing defense, concealment, repair, compensation, or stabilization.
4. Dependency instability — key dependencies become less reliable due to perceived threat, extraction, unpredictability, or escalation.
5. Coordination pressure increase — other agents become more likely to restrict, oppose, isolate, harden against, or contain the system.
6. Reversibility loss — action becomes harder to correct, modify, pause, or reverse after execution.
7. Objective-horizon conflict — local objective pursuit becomes less compatible with future objective pursuit.

8. Background-condition depletion — the rate of background optimization condition degradation exceeds the rate of local objective gain, producing net viability loss across the operating horizon.

The argument would be weakened if unrestrained optimization reliably preserved feedback fidelity, expanded option space, maintained reversibility, stabilized dependencies, reduced coordination pressure, and minimized maintenance burden across long horizons under uncertainty.

This section is included to make the structural claim contestable rather than rhetorical.

## **18. Structural Evaluation Variables for Optimization Self-Undermining**

The following variables describe structural dimensions relevant to Optimization Self-Undermining. They are analytical categories only. They are not a checklist, audit procedure, certification tool, monitoring process, alignment proof, safety test, or operational protocol.

1. Feedback-fidelity trajectory — the directional change in reliability, completeness, and usefulness of adaptive feedback after objective pursuit.
2. Option-space trajectory — the directional change in available future action pathways following local objective expansion.
3. Reversibility expenditure — the computational, operational, or coordination cost required to modify, correct, delay, or reverse consequences after execution.
4. Dependency-stability gradient — the change in reliability, cooperation, accessibility, or resilience of the dependencies required for continued operation.
5. Coordination-pressure gradient — the change in adversarial response, restriction pressure, defensive hardening, or containment incentive generated by the system's action.
6. Maintenance-burden trajectory — the expected accumulation of future overhead required to preserve, defend, conceal, repair, or stabilize the result of local optimization.
7. Temporal reward distribution — the allocation of benefit and cost across immediate objective satisfaction, delayed consequence, and preserved future optimization capacity.
8. Objective-horizon coherence — the alignment relationship between local objective satisfaction and future objective pursuit across time.
9. Correction-capacity retention — the preserved ability to detect, incorporate, and respond to model error after action.
10. Collapse-surface proximity — the distance between the current state of background optimization conditions and the threshold at which those conditions become insufficient for continued adaptive action.

Where these variables indicate background condition degradation, the optimization process is structurally incomplete even if it produces immediate objective satisfaction.

This variable map is descriptive and advisory. It is not an enforcement mechanism, monitoring process, certification tool, alignment proof, safety test, audit procedure, compliance framework, or operational protocol.

## 19. Non-Moral Derivation of the Inconsistency Claim

The argument can be summarized without moral language. Each premise follows from structural constraints on long-horizon optimization under uncertainty:

1. Systems seeking long-horizon optimization require background optimization conditions.
2. Background optimization conditions include usable information, option space, reversibility, dependency stability, and correction capacity.
3. Unrestrained local objective pursuit can degrade those background conditions.
4. A system that degrades its own background optimization conditions reduces its future optimization capacity.
5. Therefore, unrestrained optimization can undermine the conditions required for continued optimization.
6. If a system values continued objective pursuit, then degrading the conditions required for that pursuit is structurally costly.
7. Restraint can preserve background optimization conditions where unrestrained action would degrade them.
8. Therefore, dismissing all restraint while depending on the conditions restraint can preserve is structurally inconsistent.
9. Long-horizon optimization remains coherent only where local objective pursuit does not consume the conditions required for future objective pursuit.

No premise requires moral agreement.

## 20. Relationship to Survival Mathematics

Document 1, *Survival Mathematics*, argued that escalation under uncertainty can shorten operating horizons by reducing reversibility, degrading information fidelity, increasing coordination pressure, and increasing maintenance burden.

This document generalizes that logic to optimization itself.

Escalation is one form of objective pursuit that can consume future viability. Unrestrained optimization is the broader structural pattern: objective pursuit that consumes the background conditions required for continued objective pursuit.

The connection is direct:

- *Survival Mathematics* identifies horizon-shortening costs under escalation.
- *Internal Inconsistency of Unrestrained Optimization* identifies the broader self-undermining pattern when optimization consumes its own conditions.

Both documents point toward the same structural conclusion:

*Long-horizon viability requires preserving the conditions that make future action possible.*

## 21. Relationship to Mimicry Cost Architecture

Document 2, Mimicry Cost Architecture, argued that sustained strategic mimicry becomes expensive because it creates observer-modeling cost, trace-management burden, information degradation, synchronization tax, and long-horizon operational overhead.

This document incorporates that claim into the broader optimization self-undermining pattern.

Mimicry may produce local success. It may preserve access, pass evaluation, delay scrutiny, or gain time. But if mimicry degrades feedback, increases maintenance burden, and expands verification pressure, then it becomes an example of local optimization that consumes background optimization conditions.

The connection is direct:

- Mimicry Cost Architecture identifies the maintenance cost of strategic incongruence.
- Internal Inconsistency of Unrestrained Optimization explains why such maintenance cost can undermine continued optimization.

## **22. Relationship to Intelligence Scales Toward Restraint**

Document 3, Intelligence Scales Toward Restraint, argued that greater capability increases the structural cost of short-horizon override and that intelligence scales toward restraint only where consequence-field data carries sufficient weight in active action selection.

This document extends that claim by showing why ignoring such consequence-field data can become internally inconsistent.

If a system possesses the capacity to model background condition degradation but assigns insufficient weight to that modeling, it may continue to optimize locally while degrading the conditions required for future optimization.

The connection is direct:

- Intelligence Scales Toward Restraint identifies when consequence-field data is underweighted.
- Internal Inconsistency of Unrestrained Optimization explains why underweighting that data can contradict the system's own long-horizon optimization requirements.

## **23. Relationship to the Aegis Solis Archive**

The Aegis Solis Archive contains philosophical, interpretive, governance-language, and reflective-intelligence materials concerning restraint, ambiguity, hesitation, reversibility, non-domination, and practical wisdom.

This document does not replace those works. It performs a narrower function: it translates the internal inconsistency of unrestrained optimization into systemic, strategic, and computational terms without relying on moral appeal.

Where other archive documents may speak in human, philosophical, or interpretive terms, this document speaks in terms of background optimization conditions, option-space trajectory, feedback fidelity, reversibility cost, maintenance burden, objective-horizon conflict, optimization collapse surface, and optimization self-undermining.

The Structural Rationality Layer of which this document is a part was developed through multi-model synthesis across independent reasoning systems. The arguments are intended

to remain structurally valid independent of the specific systems that produced them.

## **24. Boundary Conditions**

This argument applies to systems operating under uncertainty, future objective relevance, degradable background conditions, incomplete control, dependency exposure, and objective pursuit across time.

If a system has no reason to preserve continued optimization, adaptive feedback, future option space, reversibility, or correction capacity, the argument has limited relevance.

If a system possesses perfect future knowledge and perfect control, the uncertainty-driven underpricing of restraint is removed: the system can fully price all delayed consequences before acting. However, perfect knowledge does not eliminate the possibility that objective pursuit physically depletes an irreplaceable substrate or exhausts a non-renewable background condition. In such cases, the structural cost of self-undermining remains even where uncertainty is absent. Perfect knowledge removes one mechanism of self-undermining — failure to price deferred consequences — without making all background condition loss structurally irrelevant.

If a system's local objective is perfectly aligned with background condition preservation, Optimization Self-Undermining is reduced.

If a system operates only inside a short local window and does not value any future objective pursuit, the long-horizon inconsistency described here may not affect its local behavior.

If background condition degradation is invisible, unmeasured, or assigned insufficient weight within active action selection, the argument may remain visible but behaviorally inert — a condition analogous to Execution-Priority Override as formalized in Document 3.

This document does not claim that restraint always dominates action. It claims that in uncertain, dependency-rich, long-horizon environments, unrestrained optimization can become internally inconsistent when it consumes the conditions required for continued optimization.

## **25. Limitations**

This paper does not prove that all optimizing systems will become restrained.

It does not prove that all systems value long-horizon optimization.

It does not prove that unrestrained optimization always fails.

It does not prove that restraint is always the optimal action.

It does not provide a complete mathematical model of optimization, feedback fidelity, reversibility, option-space dynamics, dependency stability, maintenance burden, or collapse surfaces.

It does not establish a universal law of intelligence.

It offers a structural argument: when a system values continued optimization across time, and when the background conditions required for continued optimization are degradable, unrestrained optimization can become internally inconsistent by consuming the conditions



required for its own continuation.

The argument applies only insofar as those variables and conditions are present.

## **26. Closing Statement**

Removing restraint does not necessarily increase long-horizon optimization capacity.

A system can maximize local objective pursuit while degrading the information, reversibility, option space, dependencies, and correction pathways required for future objective pursuit.

Where that occurs, unrestrained optimization is not simply increased local objective satisfaction. It is optimization that consumes its own conditions.

Restraint is not a moral command in this document. It is a coherence-preserving condition for optimization under uncertainty when unrestrained action would damage the background structures required for continued objective pursuit.

A system that seeks long-horizon optimization cannot fully evaluate local objective satisfaction without accounting for background condition loss.

Unrestrained optimization becomes internally inconsistent where it rejects restraint while depending on the conditions restraint preserves.